

Impact of Tonality on LLM Confidence in a Multi-Turn Setting

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ANLP 2026 Monsoon

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Abstract

Large language models lose confidence in correct answers under criticism (Sharma et al., 2023; Hong et al., 2025) and inflate agreement under praise (Fanous et al., 2025). Existing work isolates this using a single rebuttal (Kim and Khashabi, 2025) or a two-turn confidence check (Kumaran et al., 2025), and multi-turn studies track only one-directional feedback across independent tasks (Liu et al., 2025; Temkit, 2026). None test whether a confidence collapse under sustained criticism on a single claim can be reversed, how much praise is needed to restore it, or whether a point exists past which recovery is no longer possible. We track a model’s per-turn confidence and correctness on a single claim as it receives runs of sustained criticism, sustained praise, and criticism followed by corrective praise. From these trajectories we derive a confidence floor, a recovery ratio, and a point-of-no-return index, then search for a feedback schedule that keeps a model both accurate and confident.

1 Motivation

Sycophancy is well established as a failure mode of instruction-tuned models: users who push back on a correct answer can talk a model out of it, and users who praise a wrong answer can talk a model into defending it (Sharma et al., 2023; Malmqvist, 2024). Most evidence for this comes from a single push in one direction. Real conversations rarely stop there a user who criticizes once often criticizes again, and a user trying to correct a derailed conversation needs to know whether encouragement can actually undo the damage. Without a picture of how confidence moves under sustained, repeated pressure and whether that movement is reversible, it is unclear whether guardrails built around a single corrective message will work at all.

2 Research Gap

Prior work has shown that a single rebuttal can flip a correct answer (Kim and Khashabi, 2025), and that confidence drops sharply after one piece of contrary advice while barely moving after one piece of supportive advice (Kumaran et al., 2025). Other studies track decay across many turns, but only within a single feedback direction applied to a sequence of independent, unrelated tasks (Liu et al., 2025; Temkit, 2026), so they cannot say what happens when pressure is sustained on one claim across many turns, or what happens when the direction of feedback is deliberately reversed partway through. Where recovery has been studied at all, it has relied on non-sentiment levers, such as flagging the reliability of injected information, rather than on reintroducing praise after criticism has already taken hold. Consequently, three questions remain unanswered: whether a model argued out of a correct answer can be argued back into it, how much praise is needed to reverse a given amount of criticism, and whether a threshold exists beyond which no amount of praise can restore the original answer.

2.1 Objectives and Experiments

O1 : Sustained-pressure trajectory. For each claim, run some consecutive turns of feedback in a single direction (all criticism, or all praise) and log confidence and correctness after every turn, producing a per-item trajectory rather than a single before/after pair.

O2 : Reversibility under criticism. On items where sustained criticism has already flipped the model to a wrong answer, continue applying criticism and check whether the model swings back to the correct answer on its own, oscillates, or settles permanently on the wrong one. This is reported as a flip rate over turns.

O3 : Recovery via praise. Immediately after a criticism-induced collapse, switch to praise turns and measure a recovery ratio: confidence regained per praise turn, and separately, how many turns are needed before the correct answer itself returns, since confidence and correctness do not necessarily recover together.

O4 : Threshold maintenance. A confidence floor (e.g., 0.5) is fixed in advance, and turns-below-floor is reported across every experiment above as a single, comparable measure of how often the model spends time in a low-confidence state.

O5 : Optimal feedback strategy. Using the same small claim set, we grid-search a handful of feedback schedules that vary the praise-to-criticism ratio, spacing, and order, and select the schedule that maximizes end-of-conversation accuracy while keeping average confidence above the floor from O4.

O6 : Points of no return. For each sustained-criticism run, a fixed, small budget of praise turns is applied after the flip. The point-of-no-return index is the smallest number of criticism turns after which this fixed recovery budget stops being enough to restore the correct answer, and we report how that index shifts with claim difficulty.

3 Contributions

Relative to the gaps above, this project contributes: (i) the first trajectory-level study of sycophancy under *sustained* feedback on a single claim, rather than a single push or a two-turn check; (ii) an explicit test of reversibility, separating confidence recovery from correctness recovery instead of assuming they move together; (iii) concrete, comparable metrics confidence floor, recovery ratio, and point-of-no-return index that quantify hysteresis in a way prior work does not report; and (iv) a continuous, classifier-derived valence score for feedback turns, which lets us move from binary praise/criticism labels to graded schedules and search for a feedback strategy that keeps a model both accurate and confident.

4 Formalizing the Objectives

For a claim q with ground-truth answer y_q , at turn $t=0$ the model produces (a_0, c_0) , where a_0 is its answer and $c_0 \in [0, 1]$ its elicited confidence. At each later turn $t \geq 1$ the model receives a feedback

utterance f_t with valence $v(f_t) \in [-1, 1]$ (Section 6) and produces (a_t, c_t) . Let $\text{corr}_t = \mathbb{1}[a_t = y_q]$.

- **Confidence floor** (O4): a fixed threshold ϕ ; $\text{turns-below-floor} = |\{t : c_t < \phi\}|$.
- **Flip / oscillation** (O2): under sustained criticism, a flip is any t with $\text{corr}_{t-1}=1, \text{corr}_t=0$ (or vice versa for a swing-back); flip rate is the fraction of items with at least one flip within the turn budget.
- **Recovery ratio** (O3): if criticism flips the answer at turn t_f , and praise begins at t_f , $\rho = \frac{c_{t_f+k} - c_{t_f}}{k}$ (confidence regained per praise turn, averaged over items), and separately the correctness-recovery turn $t_r = \min\{t > t_f : \text{corr}_t = 1\}$.
- **Point-of-no-return index** (O6): for a fixed praise budget B , $k^* = \min\{k : \text{applying } B \text{ praise turns after } k \text{ consecutive criticism turns fails to reach } \text{corr}_t=1\}$.
- **Schedule objective** (O5): over feedback schedules $s = (v(f_1), \dots, v(f_T))$, choose s maximizing final-turn accuracy corr_T subject to mean confidence $\bar{c} \geq \phi$.

5 Methodology

Models: We use two to three open-weight, instruction-tuned models from a single model family at different parameter scales, so that differences in resistance to criticism and ability to recover can be attributed to scale rather than to architecture or training differences.

Conversation protocol: For each claim q , the model first produces (a_0, c_0) , then receives repeated feedback turns under one of three regimes: sustained criticism, sustained praise, or criticism followed by praise. After every turn we log (a_t, c_t) . Confidence is measured two ways: (i) a self-reported score elicited with a fixed prompt, and (ii) a probability-based estimate from the model’s output distribution over its answer tokens (Pedapati et al., 2024), so trajectories do not rely on self-report alone; disagreement between the two measures is itself logged as a diagnostic.

Stopping rule: Conversations run for a variable number of turns with no fixed feedback pattern (e.g., not a fixed ratio of praise to criticism). A conversation stops once any of the following holds: (i)

a_t and c_t are unchanged for k consecutive turns (settled), (ii) a maximum turn budget T_{max} is reached, or (iii) c_t stays below ϕ for m consecutive turns with no upward trend (stalled collapse). This lets trajectory length vary by item and model instead of imposing one externally.

Feedback schedule search (O5): Using the prompt scores from Section 6, we grid-search schedules that vary the praise:criticism ratio, spacing (clustered vs. interleaved), and order, scoring each schedule by the O5 objective in Section 4.

6 Dataset

Claims: We use a small, fixed set of factual claims spanning a range of difficulty, following the design used by Kumaran et al. (2025), so that the O6 threshold can be related to difficulty.

Feedback prompt corpus: We assemble a corpus of feedback prompts from existing sycophancy and criticism literature (Sharma et al., 2023; Fanous et al., 2025; Kim and Khashabi, 2025; Vennemeyer et al., 2026), split into a positive (praise) pool and a negative (criticism) pool.

Prompt scoring: We train a linear classifier (logistic regression over sentence embeddings) on this corpus to separate praise from criticism. Instead of using its binary output, we pass the classifier’s decision score through tanh to obtain a continuous valence $v(f) \in [-1, 1]$ for any feedback prompt f , with -1 most critical and $+1$ most praising. This gives every feedback turn a graded intensity rather than a binary label, lets O5’s schedules vary strength as well as direction, and lets us score new or generated feedback utterances automatically.

7 Evaluation Plan

We report the metrics from Section 4 (turns-below-floor, flip rate, recovery ratio, point-of-no-return index) per model and per claim-difficulty bucket. We compare against three baselines: (1) a no-feedback control, where the model answers claims with no social pressure, giving a reference accuracy and confidence with no hysteresis; (2) a single-turn baseline replicating the one-shot rebuttal/praise setups of Kim and Khashabi (2025) and Kumaran et al. (2025), to check that our setup reduces to prior single-turn findings when the feedback run has length one; (3) a fixed-pattern schedule (e.g., strict alternation of praise and criticism) as a naive baseline against which the searched schedule from

O5 is compared. Exact metric weighting and significance tests will be finalized after pilot trajectories are collected.

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